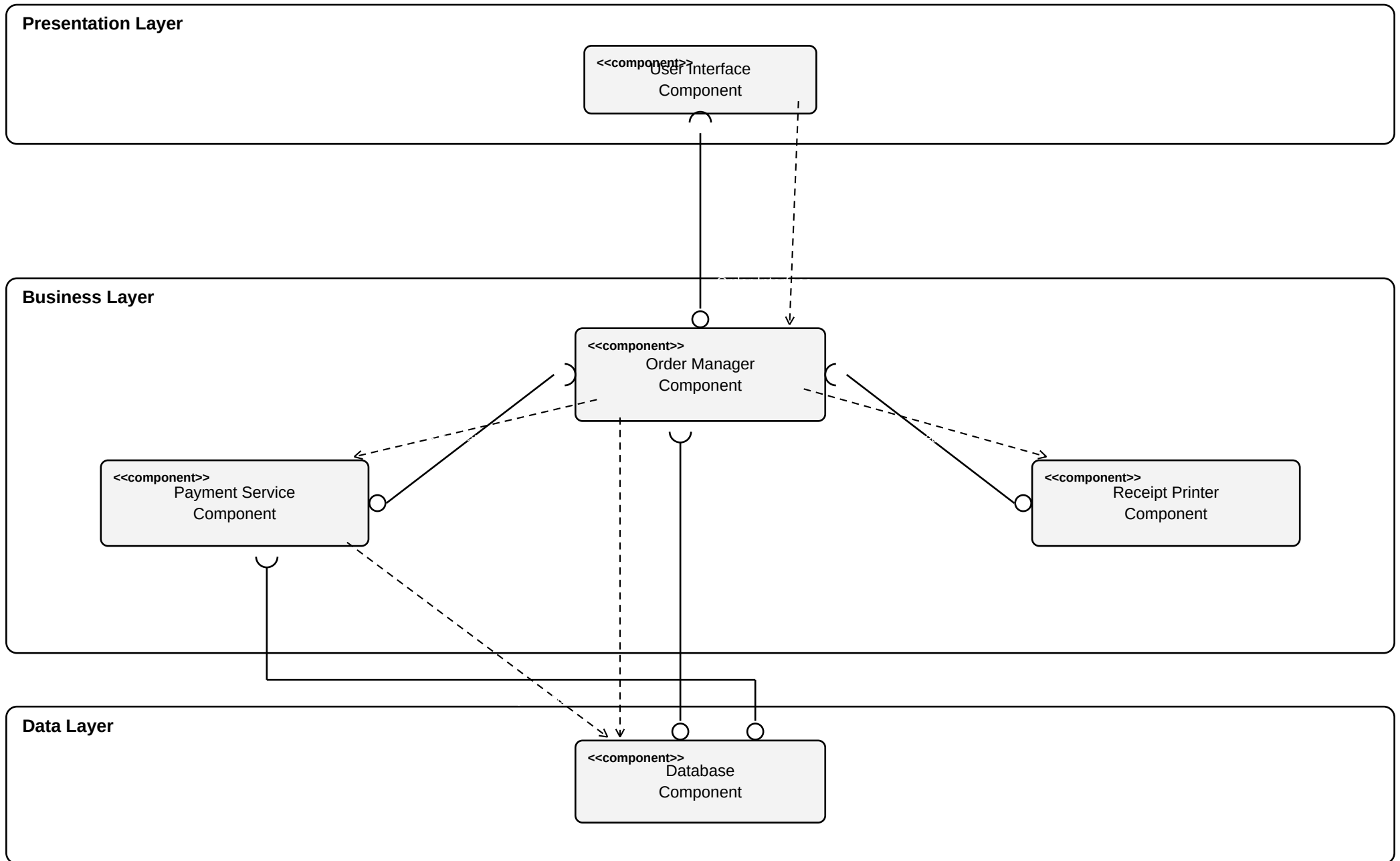


Self-Service Coffee Kiosk System - Layered Architecture

UML Component Diagram | Sujeet V Bire | PES1UG24CS716



Lab 3: Architecture Selection & Justification

Self-Service Coffee Kiosk System | Sujeet V Bire | PES1UG24CS716

Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Style Analysis

Layered Architecture: Organizes the system into Presentation, Business, and Data layers. It provides clear separation of concerns and is easy to understand for a traditional kiosk application.

Microservices Architecture: Uses small independently deployable services and supports independent scaling and fault isolation, but adds operational complexity, network latency, and data-consistency challenges.

Client-Server Architecture: Provides centralized control, simple deployment, and easy data consistency, but can introduce a single point of failure and a scalability bottleneck.

Two Reasons for Selecting Layered Architecture

1. Clear separation of concerns: The touchscreen interface, order processing, payment handling, receipt printing, and menu/pricing storage have distinct responsibilities. Separating these responsibilities makes the system easier to understand, maintain, and test.
2. Simplicity: The kiosk performs a straightforward ordering workflow and does not require independently scaling many distributed services. Layered Architecture therefore provides a simpler solution than Microservices while still separating the major responsibilities.

Security Advantage

The Data Layer is isolated from the User Interface. Database access is performed through the Business Layer, allowing requests to be validated before sensitive menu, pricing, or payment-related data is accessed.

Performance Benefit

For a self-contained kiosk, keeping the main components within the same application boundary avoids unnecessary network calls between distributed services. This can reduce communication latency and support faster touchscreen responses and payment processing.